

Adding Climate Vulnerability to Real Estate Models: A Comparative Machine Learning Analysis of Traditional Versus Climate-Enhanced Property Valuation

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Abstract

This project examines whether comprehensive climate vulnerability data improve real estate price predictions compared to traditional valuation models. While existing research primarily focuses on single climate hazards, such as flooding, I incorporated multi-hazard and baseline vulnerability data from the U.S. Climate Vulnerability Index to assess whether broader climate metrics enhance predictive accuracy. I developed and compared three machine learning models, including elastic net, random forest, and XGBoost. The project focused on three U.S. Cities – Milwaukee, Houston, and Denver. This allowed me to concentrate on neighborhood-level predictions, which provide the most benefit to individual homebuyers, local real estate professionals, and local municipalities. These cities allowed for comparison of locations with varying climate vulnerabilities and hazards.

First, I built baseline models using traditional real estate variables. Then, I created models with climate vulnerability metrics and compared model performance and predictive accuracy. Results demonstrated that climate data significantly improve predictive performance in Milwaukee, lowering RMSE and improving R-squared by approximately 0.04. However, results from Houston and Denver generated inconclusive findings.

This research contributes to understanding how comprehensive climate risk affects property values. The findings suggest that integrating climate vulnerability data could strengthen real estate valuation models and inform investment and policy decisions at the local level in certain locations. However, further research with more data is necessary to develop more robust models that are adapted more precisely to each specific location.

Problem Definition

There is a wealth of data about how the climate is changing. We also know that the effects of climate change manifest differently and to varying extents in different locations, from flooding risks in coastal areas to wildfire threats in forested regions (U.S. Environmental Protection Agency, n.d.). These hazards directly impact livability and community viability. Real estate markets are always evolving, and their changes affect almost everyone. Traditional models that are used when assessing and predicting home values include things like square footage, condition, distance to city center, school district rating, crime rates, and mortgage rates. With the data we have about how climate change is likely to affect life in different locations, there is an opportunity to determine if adding climate metrics could improve our ability to predict property values.

While a link between climate change and declining real estate values has been established (Contat et al., 2024), much of the research takes a broader, national-level perspective (First Street, 2025) or looks

only at specific climate hazards such as flooding (Gourevitch et al., 2023). The neighborhood-level insights in this report offer additional, more actionable value to local property owners and real estate professionals

Data Collection, Cleaning, and Preprocessing

As I mentioned, much of the work previously done that associates real estate values with climate change focuses on specific hazards. So that I could include more comprehensive climate change data in my final models, I used the U.S. Climate Vulnerability Index (CVI) (Tee Lewis et al., 2023). The Climate Vulnerability Index provides access to a detailed dataset that includes vulnerability measurements across multiple criteria at the census tract level. The dataset contains variables ranging from specific climate risks like temperature, drought, and precipitation. It also includes other vulnerabilities a location may have, like infrastructure, health, and socioeconomic risks. These metrics are then used in the calculation of a single number representing each location's overall climate vulnerability. For the baseline real estate data, I accessed records from the U.S. Census Bureau (2023) as well as a large dataset on Kaggle that was compiled from Realtor.com (Sakib, 2024).

My first step was to read in the Climate Vulnerability Index data from an Excel file and the Kaggle dataset from a CSV file. While the data contained in the CVI and the U.S. Census already exist at the census tract level, the properties in the Kaggle dataset are identified by ZIP code. My next step involved linking each ZIP code with an associated census tract so all the data could be merged at the same geographic level. Additionally, because the Kaggle real estate dataset contains property-specific characteristics, I aggregated the information in a way that I believed would provide the most meaning. From this dataset, I ended up using the following variables:

- modal_bed – modal number of bedrooms in each census tract,
- modal_bath – modal number of bathrooms,
- median_acres – median total acreage of the properties, and
- median_sq_ft – median square footage of the home.

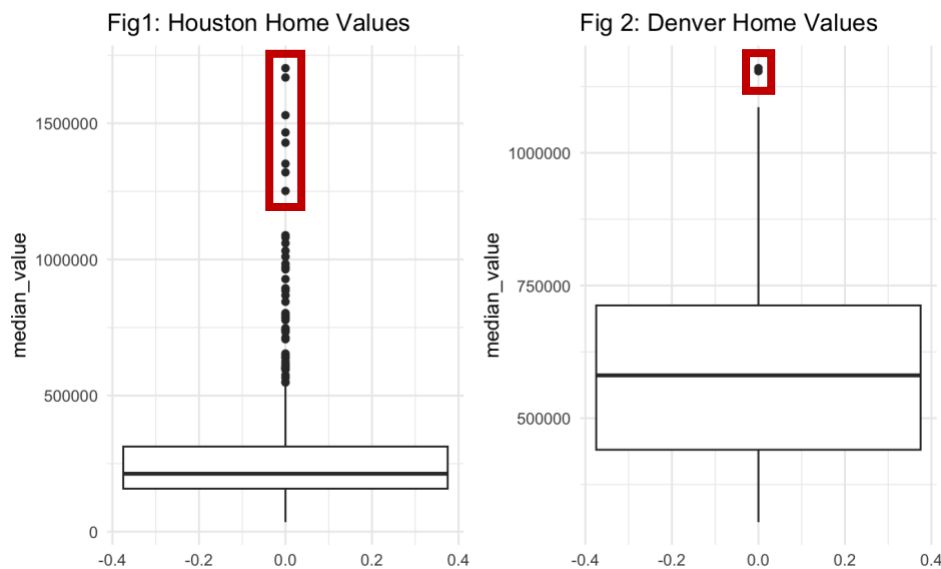
Next, I retrieved census data using the tidycensus library in R, including median income, total population, number unemployed, labor force, and median year built. From tidycensus, I also obtained each tract's median property value, which is the response variable for each of my models.

Finally, I merged data from these sources and created two separate data frames for each city – one with only traditional real estate data and one that contains climate data, including the overall Climate Vulnerability Index score as well as data on extreme climate events, such as high temperatures, flooding, and drought. Some of the climate event measurement columns contained NA values to indicate that nothing was measured, so I converted all of these values to 0, as this is worthwhile information to include in the models.

With my data frames completed, I then addressed missing values and outliers. Interestingly, many of the missing values in my data were in either the walkability or bikability columns. I decided to create a new variable that takes the average of these two scores. If, for example, walkability is missing, the new walk_bikability score will simply take the bikability score. Other missing values occurred when I merged datasets. There were some tracts listed in the U.S. Census that were not included in the CVI dataset. Because this project relies on the data in the CVI, I chose to delete these rows. Additionally, some tracts had no median home value listed. As this is the response variable, I disregarded these rows

as well. The missing data amount to 5% or less of the total. This amount is generally accepted as safe to delete with minimal effect.

Figures 1 and 2 below show boxplots of median home values in Houston and Denver. You can see values above 1250000 in Houston and 1500000 in Denver stand apart from the rest. I deemed it appropriate to remove these rows to mitigate any skew they might cause in the final models. There were no outliers significant enough to remove from the Milwaukee dataset.



Moreover, there were a handful of additional missing values in the `median_year_built` and `median_income` columns, which I imputed using median imputation so that I would not lose the data contained in the rest of the row.

Exploratory Data Analysis and Feature Engineering

With the data retrieved, organized, and cleaned, I then moved on to exploratory analysis and feature engineering. Using some of the raw variables, I created new ones that are more meaningful for this real estate application.

- First, I used the longitude and latitude coordinates to calculate the distance to the city center using central business district coordinates for each city.
- By using spatial geometry data, I calculated the area of each tract to divide it by the population to get population density.
- Next, because the census data did not directly have the unemployment rate, I derived that by dividing the number unemployed by the total labor force.
- Finally, as I mentioned in the previous section, I combined walkability and bikability scores into a single average of both to account for missing values as well as reduce multicollinearity.

Figure 3 below shows a simple glance at the statistics of some of the variables. As you will notice, the data are on vastly different scales. For example, we have median income in the tens and hundreds of thousands, and the distance to city center is no greater than 13. Both the linear

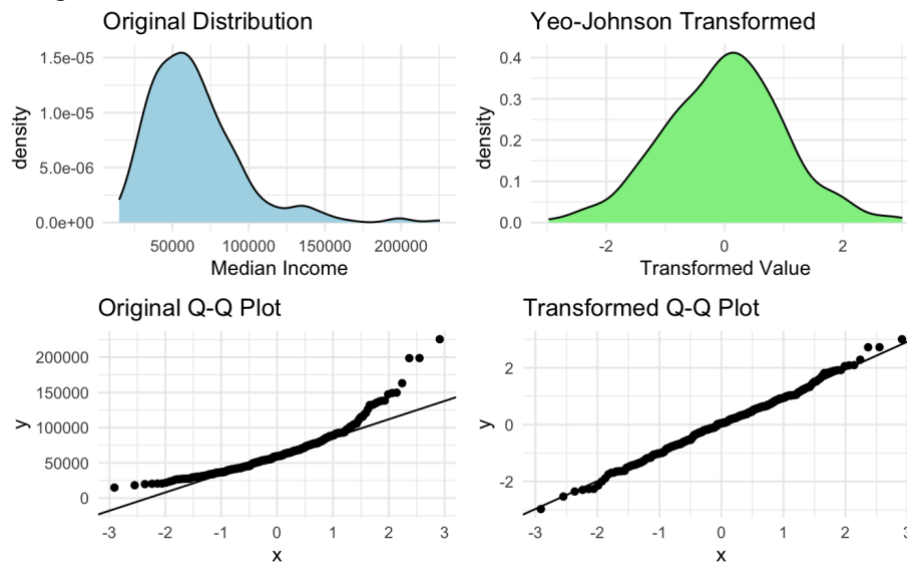
regression and penalized regression models I created required both scaling and normalization for the most accurate predictions. This is to ensure that no one predictor contributes disproportionately to the model.

Figure 3

	min	max	mean	sd	median	range	skew
Walk_Bikability	14.00	92.50	56.12	17.02	56.50	78.50	-0.14
distanceCBD_miles	0.29	12.85	5.18	2.67	4.89	12.56	0.51
median_value	36300.00	755100.00	200706.83	108925.78	183600.00	718800.00	1.73
median_year_built	1938.00	2010.00	1952.77	14.48	1951.00	72.00	0.92
median_income	14960.00	225469.00	64356.94	31065.33	59290.50	210509.00	1.65
pop_density	308.35	29914.95	7618.21	5059.48	6578.53	29606.60	1.43
unemployment_rate	0.00	0.27	0.05	0.05	0.04	0.27	1.53
modal_bed	2.00	4.00	3.17	0.52	3.00	2.00	0.22
modal_bath	2.00	3.00	2.01	0.08	2.00	1.00	11.60
median_acres	0.08	0.89	0.15	0.10	0.12	0.80	3.55
median_sq_ft	1268.50	2403.00	1649.26	256.70	1574.00	1134.50	0.82

The skew values in the table above also indicated that several of the variables were not normally distributed. Skewness between -1 and 1 is generally considered acceptable, with values closer to 0 being most ideal to satisfy normality in linear models. To ensure variables were scaled and normalized, I applied scaling, centering, and Yeo-Johnson transformation within R's caret library `train()` function using the `preProcess` parameter. See Figure 4 below for a visualization of how this affects one of the variables.

Figure 4



This visual shows the distribution and quantile-quantile plots of median_income before and after performing Yeo-Johnson transformation. You can see on the top left, median income has a right skew, and on the right, there is a more symmetrical curve. Similarly, the Q-Q plot on the bottom right shows improvement with most points lying on the diagonal compared to the original Q-Q plot on the left, indicating a more normal distribution with fewer outliers.

While these transformations are necessary for some models, they were not needed for my decision tree-based models.

Figure 5 gives a look at some of the climate variables I considered including in my models. I looked at the same statistics for these variables as I showed earlier, but the main thing I want to point out here is the column that shows the number of distinct values. This is from a data frame that is filtered for just Milwaukee, and it makes sense that some of these variables have no or very few values. Milwaukee is not located on a coast, so there would not be data for coastal flooding and sea level rise. Additionally, it makes sense that there would be fewer values for measurements like temperature since these would likely not vary much from one part of a city to another adjacent tract. As a result of these observations, I removed the following predictors from the Denver and Milwaukee datasets: Sea Level Rise, Coastal Flooding – Annualized Frequency, and Hurricane – Annualized Frequency.

Figure 5

	min	max	n_distinct
Days with maximum temperature above 35 C	10.35	13.11	53
Days with maximum temperature above 40C	2.96	4.40	53
Mean temperature	3.63	3.76	53
Urban Heat Island Extreme Heat Days	12.33	14.33	7
Drought – Annualized Frequency	0.78	4.28	6
Wildfire – Annualized Frequency	0.00	0.00	136
Total Precipitation	7.24	7.51	53
Coastal Flooding – Annualized Frequency	0.00	0.00	1
Riverine Flooding – Annualized Frequency	0.00	1.88	2
Sea Level Rise	Inf	-Inf	1
Hurricane – Annualized Frequency	0.01	0.01	1
Tornado – Annualized Frequency	0.00	0.01	177
Overall CVI Score	0.31	0.55	278
Climate Change: All	0.30	0.44	278
Climate Change: Extreme Events	0.36	0.57	207

As a final step in this exploration phase, I examined variable multicollinearity. The models I built generally handle multicollinearity well and select out or reduce the importance of variables with high multicollinearity; however, I still wanted to be aware of it. I explored this in two ways. Figure 6 on the left is a correlation plot, where the darker-colored circles represent a higher correlation coefficient. The table in Figure 7 on the right shows Variance Inflation Factor (VIF) scores. Even in elastic net, VIF scores greater than 10 can cause problems, so I paid attention to the highlighted variables when I built my models. It makes sense that these are highly correlated, as the first group is based on temperature measurements. The second group is highly correlated because Climate Change: Extreme Events is a subset of Climate Change: All. In the next phase of the project, I compared model performance with these correlated variables included and models where I only selected one from each correlated group.

Figure 6

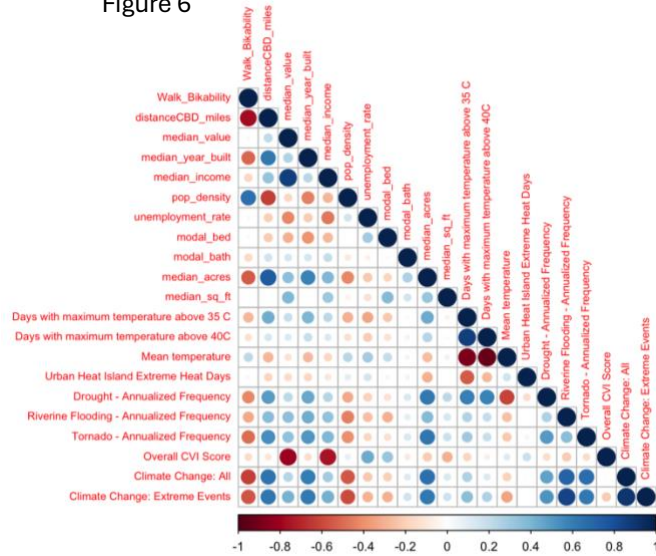


Figure 7

variable	VIF
Walk_Bikability	4.034583
distanceCBD_miles	7.369072
median_year_built	2.455598
median_income	3.765173
pop_density	2.371751
unemployment_rate	1.586045
modal_bed	2.442393
modal_bath	1.201630
median_acres	5.397817
median_sq_ft	2.543529
Days with maximum temperature above 35 C	35.434637
Days with maximum temperature above 40C	17.326873
Mean temperature	31.399424
Urban Heat Island Extreme Heat Days	5.800114
Drought - Annualized Frequency	3.285662
Riverine Flooding - Annualized Frequency	5.777985
Tornado - Annualized Frequency	2.651590
Overall CVI Score	4.215839
Climate Change: All	9.971407
Climate Change: Extreme Events	19.467843

Model Development and Evaluation

I set out to build five models and compare their predictive accuracy for data with and without climate metrics added. First, I created a simple linear model to use as a baseline to compare with more complex models. See the results in Figure 8. For each metric, the values on the left represent models with only real estate data, while the values on the right are for models that include the climate vulnerability data. RMSE here represents, on average, how much predictions for median home value differ from their actual values. We can see that there is room for improvement over this simple model.

Figure 8

City	RMSE		R ²	
Milwaukee	58,712	52,794	0.64	0.72
Houston	99,564	100,874	0.72	0.72
Denver	87,997	92,632	0.86	0.83

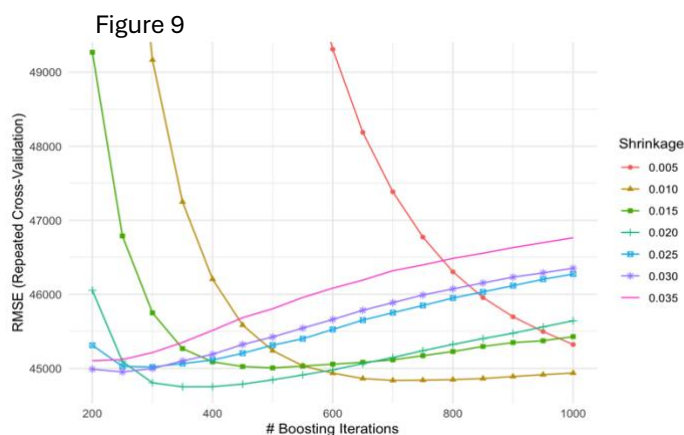
Next, I generated an elastic net (e-net) model, which is a penalized regression model that allows for variable selection within it. I also built two decision tree models, random forest and extreme gradient boosting (XGBoost). Again, because of my correlated predictors, random forest is a suitable choice because only a subset of predictors is considered at each split. This method is also considered quite interpretable, particularly in its ability to provide metrics on feature importance. While less interpretable and more complex than random forest, XGBoost is generally known to give more accurate predictions than random forest. Furthermore, research by Sharma et al. (2024) found XGBoost to be the best-performing model for housing price prediction. Finally, I attempted to build an artificial neural network (ANN) model. While notorious for being difficult for laypeople to understand, I wanted to explore the ANN's ability to handle more complex interactions among my predictors. However, in the process of tuning hyperparameters, I was unable to achieve predictive accuracy to rival the other models. Because of this and ANN's lack of interpretability as well as its high computational cost, I made the decision to abandon this model and focus my attention on the ones I already mentioned.

I tuned hyperparameters of e-net, random forest, and XGBoost through a manual iterative grid search process and applied 10-fold repeated cross-validation, starting with broad ranges of values and

narrowing them down based on which values gave the best results (lowest RMSE). For e-net I tuned the alpha and lambda values, alpha controlling the balance between LASSO and Ridge, and lambda, which controls the strength of the penalty. Random forest required the tuning of just one parameter, mtry, which is how many features are randomly sampled at each node. With seven hyperparameters, XGBoost involved the most time and complexity to tune. As part of my iterative grid search, I used plots like the one in Figure 9 to compare variations of different parameters.

You can see it shows the RMSE for different numbers of boosting iterations, with each line color representing a different learning rate. I also created a function that adds a row to a table with the values of the best tune of a particular grid search, sorted based on the RMSE, helping me keep track of the performance of different hyperparameter combinations.

This initial, iterative grid search process was the first step I took to narrow down the ranges of parameters I further tuned in the double cross-validation process. With smaller datasets of under 500 rows, 10-fold double cross-validation was possible to assess the full modeling process and determine the best predictive model for each city.



Results, Insights, and Recommendations

Based on both R^2 and RMSE, I achieved better results with more sophisticated models compared to the simple baseline linear model I started with for Milwaukee and Houston. Interestingly, XGBoost outperformed the other models for both Milwaukee and Houston, while E-net performed best with the Denver data. With results from models using only real estate data on the left and models incorporating climate data on the right, we can see in Figure 10 that adding climate data to the models does appear to improve their predictive accuracy. Notably, a paired t-test did not produce a p-value below the standard threshold of .05 for statistical significance for Houston or Denver. Therefore, we cannot say that the difference in results between the two models for these cities is significant enough to attribute to adding climate factors to the model. It is also important to mention that for Denver, none of these models achieved an improvement over the simple linear model I discussed earlier. These results are unexpected since it has been established through prior research that climate factors usually do affect home values. Results would likely improve with larger datasets, particularly for Denver, as it only had 108 rows.

Figure 10

	Best Model	RMSE		R^2		P-value
Milwaukee	XGBoost	45,213	39,826	0.83	0.87	< .001
Houston	XGBoost	92,027	90,629	0.75	0.76	0.60
Denver	E-net	111, 177	104,337	0.66	0.70	0.35

A closer look at the residuals confirms that the models perform reasonably well, but there is room for improvement. We hope for residuals that follow close to a normal distribution without any signs of unusual patterns. In Figure 11, we have a Q-Q plot, and in Figure 12, there is a histogram of residuals. I am encouraged to see that most points fall on or close to the Q-Q line. However, we do notice some outliers creating a slight skew to the right. The plot in Figure 13 displays the residuals against the model's predicted values. Here, we see a reasonably random scatter, which indicates that there are no underlying systemic errors or problems in the model. Going back and refining the original data to handle outliers better could help further reduce error in the model and give more accurate results.

Figure 11

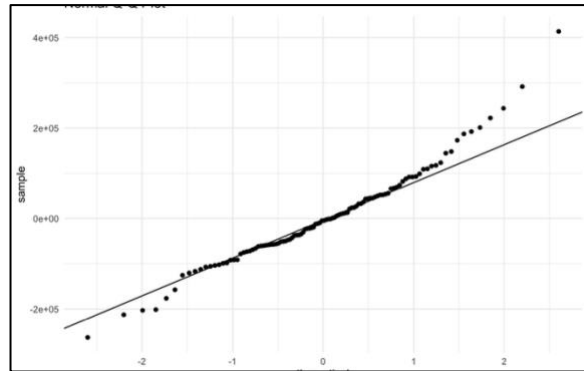


Figure 12

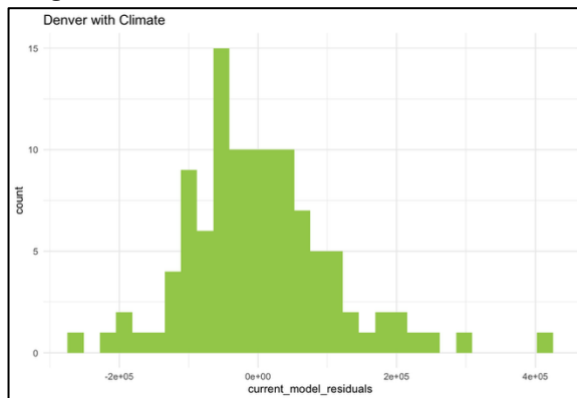
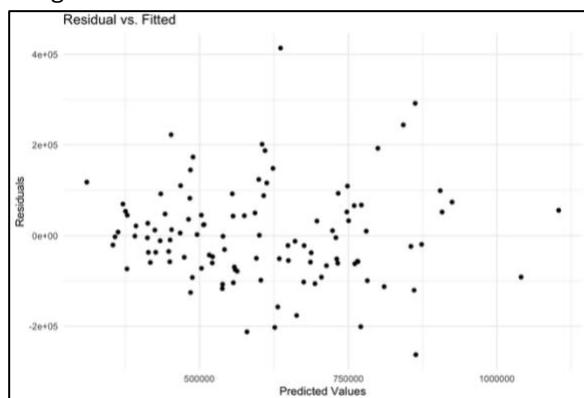
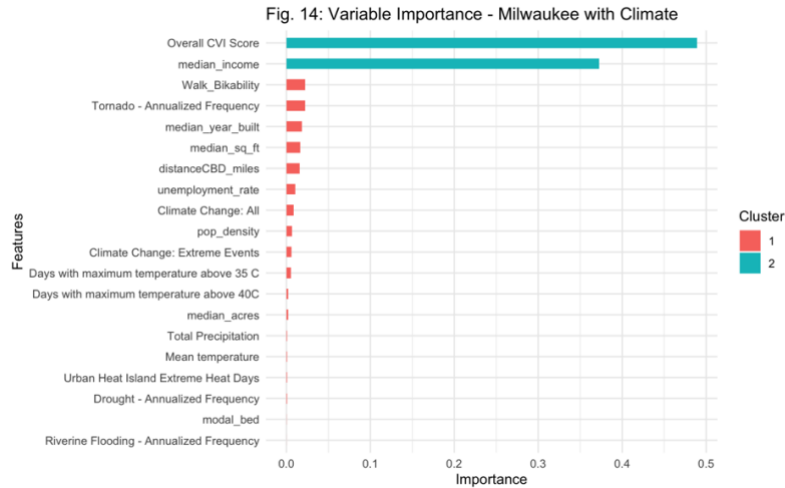


Figure 13

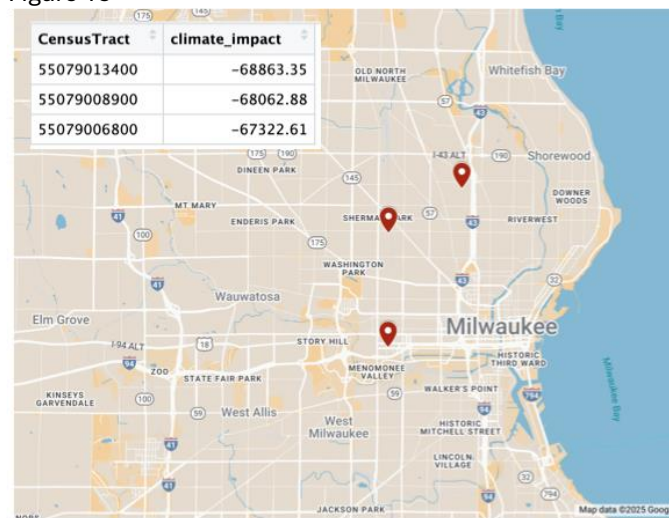


For at least Milwaukee, climate vulnerability data do appear to improve home pricing predictions. The variable importance graph in Figure 14 was generated after retuning and fitting the best model on the entire dataset. The graph illustrates that, amongst the climate data, the CVI score has the most effect on home values, more than individual climate hazards. Climate hazards that play an important role in home values are likely specific to each location, which highlights the importance of building these models for each city individually.



SHAP Analysis of the final model helped me identify the neighborhoods that are most affected by climate factors. The points on the map of Milwaukee in Figure 15 show the locations of the most vulnerable neighborhoods. The table at the top indicates how much median home value loss is predicted from the most important climate variables.

Figure 15



With confirmation that climate vulnerability data likely enhance the predictive power of real estate valuation algorithms in Milwaukee, I would make the following recommendations:

- With the focus at the neighborhood level, real estate agents and firms can better advise clients with enhanced property analysis, noting CVI scores and helping them understand the role they play in property values.
- Local governments can use predictive models like this that include CVI data to make zoning decisions, tax assessments, infrastructure investments, and housing policy, concentrating first on the most vulnerable neighborhoods identified above.
- Individual home buyers should factor climate risk into property purchasing decisions.
- Mortgage lenders should consider factoring CVI scores into their loan approval algorithms.

Conclusion

This research demonstrates that incorporating comprehensive climate vulnerability data significantly improves real estate valuation models at the neighborhood level in some locations. Analysis of Milwaukee showed that climate-enhanced models outperformed traditional approaches, with R^2 values exceeding 0.75 and improved upon baseline models by 0.04. The inconclusive Denver and Houston results highlight the need for larger datasets to draw definitive conclusions. The CVI score was the most influential climate predictor. These findings have practical implications for real estate professionals, local governments, and individual home buyers in Milwaukee.

Future work should focus on three areas: refining models by integrating additional real estate data, expanding the scope to include larger metropolitan areas rather than city boundaries alone, and applying this methodology to additional U.S. cities with diverse climate hazards and varying preparedness levels. This would strengthen the generalizability of these findings and provide a framework for climate-informed real estate valuation nationwide.

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